

Detecting the effects of management regime shifts in dynamic environments using multi-population state-space models

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ABSTRACT

Detecting the effectiveness of management actions intended to increase the abundance of threatened or exploited species can help resolve uncertainties about cost-effective management tactics. However, the complexity of ecological systems can make it difficult to identify important factors causing change in population abundance. This difficulty extends from detecting naturally-caused ecosystem regime shifts to management-induced regime shifts and the attendant change in population dynamics parameters. The adult abundance of naturally-produced coho salmon (*Oncorhynchus kisutch*) on the Oregon Coast generally declined until these fish were listed as threatened under the Endangered Species Act in 1998. The subsequent rebuilding of Oregon coastal coho abundance is coincident with increased habitat restoration, reduced hatchery production, and reduced harvest. Importantly, ocean survival also improved, thereby complicating the assessment of management effectiveness at the adult life stage. Our objective was to assess change in the freshwater production of juveniles (smolts) through time in order to determine if recent increases in adult abundance could be related to management affecting the freshwater juvenile production. We combined 46 years of data associated with 18 populations of Oregon coastal coho. Spawner-to-smolt relationships were modeled with Bayesian hierarchical state-space implementations of the logistic hockey stick recruitment function. We also develop a method of estimating the relative reproductive success of hatchery spawners. We found more evidence for decline than increase in productivity in the spawner-to-smolt life stage, suggesting that changes in physical oceanographic conditions are responsible for recent increases in adult abundance. The reproductive success of hatchery-origin fish relative to natural-origin fish was 0.51 with a 95% credible interval from 0.19 to 0.89. While some management effects may unfold on longer time-scales than we observed, we nonetheless contend that carefully tailored models of non-stationary population dynamics are needed to understand and the effectiveness of management actions intended to recover populations.

1. Introduction

The status of fish populations and associated fisheries is a paramount concern that extends from hearty academic debate to uncertainty in management actions affecting the fate of species, local economies, and food security (Szuwalski, 2016). The perceived decline in abundance of fishes around the world has induced change in management actions intended to enhance sustainability (Worm et al., 2009; King et al., 2015). The ability to detect the effects of these management actions on population dynamics is constrained by the simultaneous changes in complex ecological systems, and the models and data used to understand those systems (Loneragan, 2014).

The common assumption that intrinsic population growth rate and the density-dependent crowding effects that give rise to “carrying capacity” (Mallet, 2012) are fixed, stationary values may be useful to

parsimoniously describe a time series of observed abundances, but cannot be true over relevant ecological time scales because of change in the abiotic (Kilduff et al., 2015) and biotic (Pinsky et al., 2013) components of nature (Szuwalski et al., 2015). Thus there is a great deal of interest in detecting nonstationary population dynamics caused by these natural “regime shifts” (Vert-pre et al., 2013). We contend that both natural regime shifts and the large-scale management efforts intended to enhance sustainability- “management regime shifts”- can be studied with similar analytical methods. Detecting management regime shifts has received relatively little attention, but is nonetheless an important component of resolving uncertainty in management strategies (Punt et al., 2016).

Oregon coastal coho salmon (*Oncorhynchus kisutch*) provide a good opportunity to apply methods of detecting regime shifts on population dynamics. The adult abundance of naturally produced Oregon coastal

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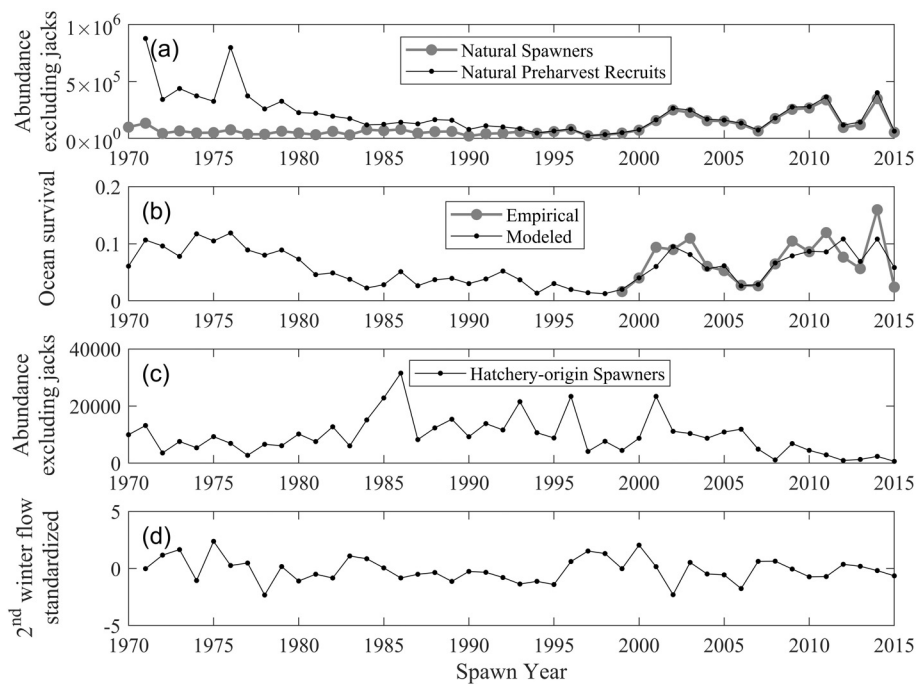


Fig. 1. The abundance of natural-origin adult coho declined until the late 1990s (a), which is coincident with the pattern in ocean survival rate (b), but multiple biologically conservative actions also began in the late 1990s, including reduction in hatchery production (c). Mean second winter stream flow from Nov–March in four sites were standardized (z-score) over 1961–2015 and averaged together (d).

coho (*O. kisutch*) salmon generally declined until they were listed as threatened under the United States federal Endangered Species Act (ESA) in the late 1990s. Following the ESA listing, adult abundance began to increase (Fig. 1a) and reached a 30-year high in 2013. This increase is coincident with two fundamentally different processes. First, fishery-independent survival in the ocean improved (Fig. 1b). Second, several large-scale management changes were implemented leading up to and following the ESA listing. These management changes include: (i) reduction in ocean and freshwater harvest that caused the harvest rate to decline from a maximum of 91% in 1976 to a minimum of 3% in 2008 (Fig. 1a), (ii) reduction in hatchery production that caused the abundance of hatchery-origin fish spawning in the wild to decline from a maximum of 31,530 in 1986 (when the average spawning population had 27% hatchery fish) to just 662 in 2015 (when the average spawning population had 0.7% hatchery fish; Fig. 1b), and (iii) freshwater restoration investments that totaled over 100 million US dollars between 1997 and 2003 (Nicholas et al., 2005), and over 106 million US dollars of state and federal funding for projects begun between 2000 and 2013 (OWEB, 2017).

Separating how much of the increase in adult abundance is attributable to improvement in natural ocean conditions versus a shift in the management regime will improve understanding of management effectiveness and long-term population viability. For example, ocean conditions are largely beyond management control, so if ocean survival rates return to the low values observed in the late 1990s (Fig. 1c), then the direction and rate of change in adult abundance will depend, in part, on how well management actions have improved spawner-to-smolt production. If management actions have improved smolt production, then populations will be more likely to persist through periods of low ocean survival.

Here, we use 46 years of monitoring data from 18 populations of Oregon coastal coho (Fig. 2) to determine whether a change in spawner-to-smolt recruitment parameters (hereafter “smolt recruitment”, which occurs in freshwater) coincides with a biologically conservative shift in the management regime. If ocean conditions, not management, are responsible for changes in adult abundance, then we expect to find no change in smolt recruitment across two periods roughly corresponding to pre- and post-biological conservation management. In contrast, if management actions have meaningfully

improved smolt recruitment, then this should be detectable in the abundance time-series data after controlling for changing ocean conditions and changing proportions of hatchery-origin spawners, which may exhibit lower reproductive success than natural-origin fish. We evaluated the evidence in data for these contrasting hypotheses by reconstructing historical ocean survival and then fitting a density-dependent life-cycle model using a hierarchical, state-space approach.

2. Methods

2.1. Abundance time-series

The abundance of coho salmon spawning in multiple populations on the Oregon Coast has been monitored by the Oregon Department of Fish and Wildlife for many decades using many methods. We use annual spawner abundance estimates from 1970 through 2015. Although earlier estimates exist, we chose 1970 as the beginning of the time series for consistency with other authors who noted that earlier estimates are relatively unreliable (Rupp et al., 2012a; Rupp et al., 2012b). Improvements to the abundance sampling design were made after 1970 (Jacobs and Nickelson, 1998; Lewis et al., 2009), so we created statistical models that permit change in the magnitude of sampling error that correspond with three different periods of sampling design: 1970 to 1990, 1991 to 1998, and 1999 to 2015. These three periods correspond to the use of index site sampling, stratified random sampling, and generalized random tessellation sampling, respectively.

The time series for the Beaver Creek population contains missing values from 1975 to 1979 so we began that population's time series 1980. There are 15 missing values of spawner abundance scattered throughout the remaining populations and years, which do not pose a problem in our subsequent Bayesian analysis because these values can be treated as unknown parameters to be estimated (Gelman and Hill, 2006). The proportion of hatchery-origin and naturally-produced spawners is also known for each population and year through analysis of carcass scales and visual inspection for adipose fin removal.

2.2. Harvest time-series

Ocean harvest rates of coho in the Oregon Production Index (OPI)



Fig. 2. Eighteen populations of coho salmon on the west coast of Oregon, USA, are defined by drainage basins. Abundance has been estimated annually since 1970 (see text for exceptions). Smolt-to-adult survival has been recorded since 1998 at sites located within Nehalem, Siletz, Yaquina, Alsea, Lower Umpqua, and Coos.

area from 1971 to 1994 were reported by the Pacific Fisheries Management Council (Salmon Technical Team, 1995). The OPI harvest estimates aggregate (i) harvest of both naturally-produced and hatchery-reared fish, (ii) populations that enter the Columbia River, which are not part of this study, with the focal populations in the Oregon coastal range (Fig. 2), and (iii) ocean and freshwater harvest. However, harvest rates of just naturally-produced Oregon coastal coho in the ocean are needed for our analysis. Such harvest rates are available from 1986 to 2014 using the fishery regulatory assessment model (FRAM) implemented for mixed stock management by the Oregon Production Index Technical Team (MSM-OPITT) of the PFMC. By regressing logit-transformed FRAM-derived harvest rates on OPI area harvest rates ($R^2 = 0.90$, $P = 0.0002$, $df = 6$), we estimated ocean harvest of naturally-produced Oregon coast coho from 1985 back to 1973, which is the earliest year of smolt recruitment from the first spawn year (1970). This procedure assumes that the relative composition of populations in the catch from 1986 through 1994 is representative of the period from 1973 through 1985. Changes to the fishery began around 1994, which should not affect the projection back in time as much as it would affect a forward projection.

Freshwater harvest rates from 1970 to 1993 are not known uniquely for each population, so an aggregate freshwater harvest rate was applied to all populations (Melcher, 2005). Beginning in 1994, freshwater harvest of Oregon coastal coho was prohibited, but we applied a 1% incidental mortality rate to coho populations that overlap a Chinook fishery (Melcher, 2005). Beginning in 2009, some coho populations experienced direct freshwater harvest in a quota-based fishery, and these harvest rates were applied to individual populations.

2.3. Ocean condition

Coho smolt-to-adult survival rates have been monitored at six sites distributed along the Oregon Coast since 1998 (Suring et al., 2015, Fig. 2). Coho smolt-to-adult survival rates were derived by dividing the adjusted female spawner abundance in year t by half of the smolt

production, assuming a 1:1 smolt sex ratio, in year t^{-1} at each site. Spawner abundance was adjusted for harvest impacts by dividing abundance by one minus the year specific naturally-produced Oregon coastal coho harvest rate. The coast-wide smolt-to-adult survival rate was calculated by averaging the marine survival rate from all sites (ODFW Oregon Department of Fish and Wildlife, 2013). Amendment 13 to the Pacific Coast Salmon Management Plan currently uses these estimates in a predictor for Oregon coast coho ocean survival (ODFW Oregon Department of Fish and Wildlife, 2007). This predictor cannot be used to infer marine survival prior to 1998 because it uses datasets not collected prior to that time, however, Rupp et al. (2012b) created a predictor of Oregon coast coho salmon abundance using data available from 1970 to present. Rupp et al. (2012b) related ocean abundance (spawning escapement plus harvest-related mortality) to seven oceanographic variables by model averaging six 3-variable generalized additive models. Rupp et al. (2012b) took precaution to avoid spurious correlations and demonstrated a good relationship between oceanic variables and ocean abundance of coho. Their model has high forecast ability (ordinary cross-validation = 0.74) of adult abundance back to 1970. We refit the six 3-variable generalized additive model after substituting logit-transformed smolt-to-adult survival for adult abundance as the response variable (Table 1). The six fitted models were averaged in logit space to produce a single inferred survival value annually back to 1970.

2.4. Freshwater condition

High and low stream flow during juveniles' second winter (juveniles are eggs during their first winter, Fig. 3a) is thought to negatively affect survival (Lawson et al., 2004). Lawson et al. (2004) used U.S. Geological Survey stream flow data recorded in the Nehalem, Wilson, Siletz, Alsea and Coquille. We were unable to find Coquille stream flow data with the temporal coverage described by Lawson et al. (2004) so we substituted it with data from the Umpqua. As in Lawson et al. (2004), mean stream flows from November through March were averaged

Table 1

Six different three-variable generalized additive models were identified by [Rupp et al. \(2012b\)](#) to predict coho ocean abundance. These models were refitted with smolt-to-adult survival as the response variable. The annual smolt-to-adult survival prediction is an average of the logit transformed values predicted by each of the six generalized additive models. Variables used in the models are four-year moving average of Pacific decadal oscillation index (PDO), date of spring transition between down-welling and upwelling (SPR), upwelling index (UWI), sea surface height (SSH), spawner abundance (SPN), Multivariate El Niño-Southern Oscillation index (MEI), and sea surface temperature (SST). The three letters following the dot give the three consecutive months over which the variable was measured. The coefficient of determination (R^2) and ordinary cross validation (OCV) are given for individual models and the model averaged ensemble.

ID	Variables			R^2	OCV
Model 1	PDO.MJJ4	SPR	SPN	0.71	0.36
Model 2	PDO.MJJ4	MEI.OND	UWI.JAS	0.57	0.07
Model 3	PDO.MJJ4	SPR	MEI.OND	0.68	0.45
Model 4	PDO.MJJ4	UWI.JAS	SST.AMJ	0.61	0.24
Model 5	PDO.MJJ4	SSH.AMJ	UWI.JAS	0.77	0.50
Model 6	PDO.MJJ4	UWI.SON	SST.J	0.74	0.46
Models 1–6 ensemble				0.75	0.47

together within each location. Each location's annual values were then standardized to a z-scale by subtracting the mean out of each value and then dividing by the standard deviation. The z-transformed values were then averaged across locations within each year ([Fig. 1d](#)).

2.5. Recruitment model

Coho salmon are semelparous, and density-dependence in population dynamics can be conveniently modeled with a function relating the abundance of spawning fish on a given year (S) to the abundance of their progeny that recruit (R) into a given life stage. [Barrowman and Myers \(2000\)](#) introduced the “logistic hockey-stick” recruitment model:

$$R = \alpha \theta \gamma (1 + e^{-1/\theta}) \left(\frac{S}{\theta \gamma} - \log \left(\frac{1 + e^{(S-\gamma)/(\theta \gamma)}}{1 + e^{-1/\theta}} \right) \right). \quad (1)$$

The parameter α is the slope at the origin in a plot of recruits over spawners, and is equivalent to the “intrinsic productivity” parameter in the Ricker and Beverton-Holt recruitment models. The parameter θ controls the smoothness of the transition to asymptotic recruitment and the parameter γ is the inflection point of the curve. We will subsequently focus on changes in γ across periods as an indication of change in the production of smolts per spawners.

[Barrowman and Myers \(2000\)](#) provide theoretical and practical justification for the logistic hockey-stick model, noting that researchers “resort” to Ricker or ad hoc nonparametric approximations because Beverton-Holt models often produce poor estimates. For the coastal coho data analyzed here, the posterior probability distribution of the α parameter in a Beverton-Holt model equaled or closely matched the prior probability distribution in about one third of the populations. This problem persisted across different prior parameterizations for α . This suggests that parameters of the Beverton-Holt model cannot be reliably estimated from these data. Although the Ricker model can yield reliable parameter estimates, the overcompensation of the Ricker model does not seem appropriate for coho salmon ([Bradford et al., 1997](#), [Barrowman and Myers, 2000](#), but see also [Buhle et al., 2009](#) for an application of a Ricker model to coastal coho).

Spawning fish include both natural-origin (“wild”) fish and hatchery-origin fish. However, hatchery-origin fish cannot be included as recruits from a previous generation of spawners because hatchery fish, in fact, come from hatcheries. Furthermore, hatchery-origin fish that spawn in the wild may produce offspring that are relatively maladapted to the natural environment ([Christie et al., 2014](#)). Hatchery-origin fish spawning in the wild may therefore produce fewer progeny that survive to adulthood than natural-origin fish. To remove the effects

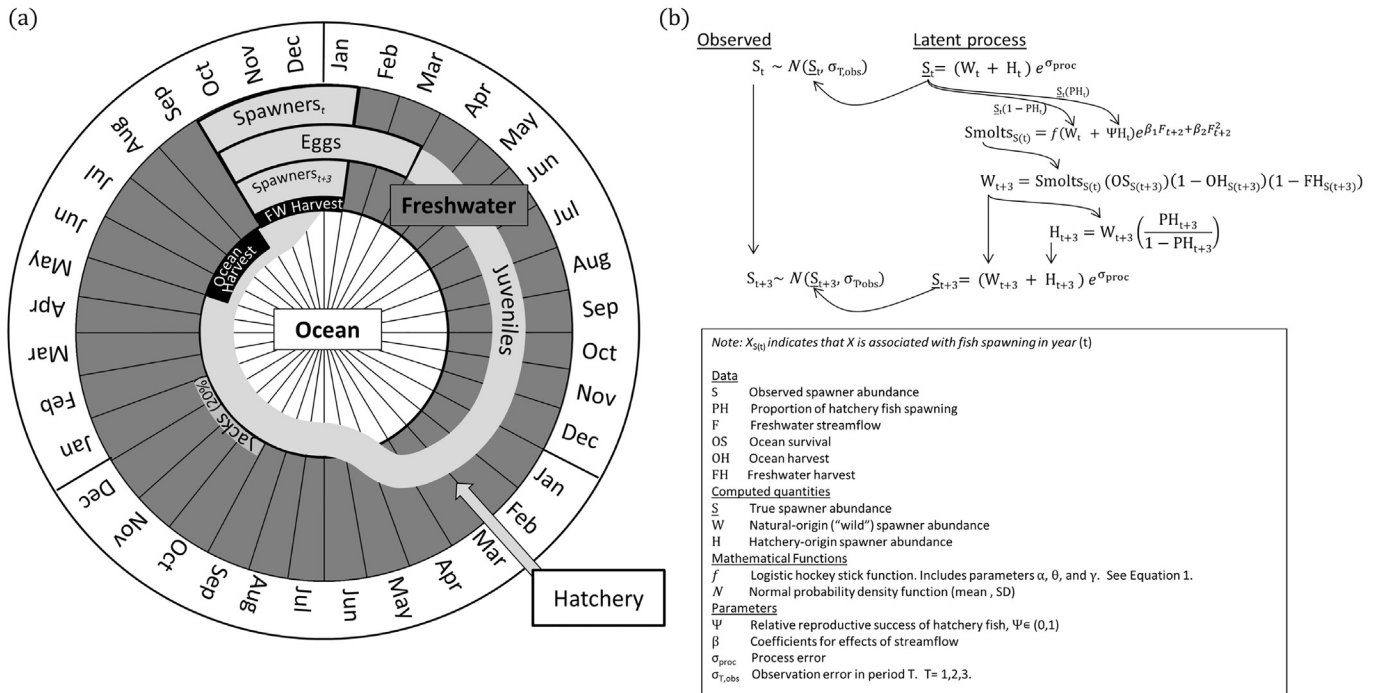


Fig. 3. Life cycle of Oregon coastal coho (a). Jacks are precocious males and not included in the data or models presented here. The state-space model (b) of coho population dynamics decomposes spawner abundance into hatchery and natural-origin components using annual estimates of the proportion of hatchery fish. Hatchery and natural-origin fish produce smolts, but reproductive success of hatchery fish is some fraction (Ψ) of natural-origin fish that can be estimated empirically. The latent process generating spawner abundance is recursive, but hatchery-origin fish are added to match the observed fraction of hatchery-origin spawners. Process error is logarithmic, reflecting the common assumption that recruitment is a multiplicative survival process. Each year's estimate of spawner abundance is assumed to be normally distributed around the true abundances, reflecting unbiased sampling error. The magnitude of the sampling error is allowed to vary across three time periods when different sampling designs were used. The parameter of the recruitment function controlling the inflection point (γ) was allowed to vary across two periods (before and after 1998) that reflect times of pre- and post-biological conservation management.

of changing proportions of hatchery-origin fish on the spawning grounds (Fig. 1c) on recruitment performance, we estimated the relative reproductive success of hatchery-origin fish within the state-space model of population dynamics (Fig. 3b).

2.6. State-space modeling

State-space modeling has been used to separate process variation from observation error in several stock-recruitment analyses (Meyer and Millar, 1999; Millar and Meyer, 2000a; Millar and Meyer, 2000b; Rivot et al., 2004; Su and Peterman, 2012). These models treat the spawner abundance data as a time-series, where a given year's abundance generates subsequent spawner abundance through a nonlinear function (i.e. the recruitment function). It is possible to write a model for the true yet latent process giving rise to the abundance time-series, and then treat abundance data as imperfect observations of that process (Fig. 3b). Importantly, the true yet imperfectly observed abundances are mechanistically connected through time by the recruitment function. Thus any deviation in true abundance on a focal year has implications for the abundances in previous and subsequent generations. In contrast, observation errors will not propagate through a time-series because there is no mechanism linking observation errors through time.

Since most male coho salmon and nearly all female coho mature as three year olds, we ignore precocious individuals and model population dynamics with a strict three-year life-cycle (Fig. 3a). Each population therefore consists of three time-series (beginning 1970, 1971, and 1972 and ending 2013, 2014, 2015, respectively). We initialized the joint time-series of true abundances (S) by setting the first three values to the observed abundances (S):

$$S_{-t=1970} = S_{t=1970}$$

$$S_{-t=1971} = S_{t=1971}$$

$$S_{-t=1972} = S_{t=1972}$$

The true spawner abundance in year t can be decomposed into hatchery-origin (H) and natural-origin (W) components using the observed proportion of hatchery fish spawning (PH) in year t . We then multiplied the number of hatchery-origin spawners by a Beta-distributed parameter (Ψ , Fig. 3b). Using a Beta distribution ensures that Ψ is confined to a number between zero and one. If $\Psi = 1$ then hatchery fish have identical reproductive success as natural-origin fish. If $\Psi = 0$ then hatchery fish do not produce any surviving offspring. Thus the parameter Ψ expresses the reproductive success of hatchery-origin fish relative to natural-origin fish for all levels of spawning escapement. To estimate Ψ from data, we introduce it into the recruitment function (Eq. (1)) by making the following substitution for spawner abundance:

$$\underline{S}_t = \underline{S}_t(1 - PH_t) + \Psi(\underline{S}_t)(PH_t).$$

The recruitment function, f , is multiplied by an exponential scale quadratic function of freshwater flow (F) during juveniles' second winter. It is customary to exponentiate covariates because survival is a multiplicative processes that yields lognormal errors in recruitment (Hilborn and Walters, 1992). Using an exponential scale for the freshwater flow covariate returns a number between zero and infinity. This is convenient because the effects of flow are expressed as a multiplier that is independent of population size. Effects of streamflow can therefore be used for all populations, which vary in overall population size, without having to uniquely estimate parameters for each population. We call the result of this computation "smolts" (Fig. 3b). To complete the life cycle, and hence mechanistically connect the time-series of abundance, we multiply smolt abundance by the natural ocean survival rate (OS) and the quantity 1 less harvest rates in the ocean (OH) and fresh water (FH) associated with spawners at $t = 3$. The result is the

number of natural-origin spawners:

$$W_{t+3} = f\left(\underline{S}_t, \Psi, \alpha, \theta, \gamma\right) (e^{\beta_1 F_{t+2} + \beta_2 F_{t+2}^2}) (OS_{t+3}) (1 - OH_{t+3}) (1 - FH_{t+3})$$

where f is the smolt recruitment function (Eq. (1)). To statistically relate W_{t+3} to the observed spawner abundance in year $t + 3$, we must add back hatchery-origin fish and include lognormal process variation (σ_{proc}) in the life-cycle recruitment process:

$$\underline{S}_{t+3} \sim \text{LogNormal}\left(W_{t+3} + W_{t+3} \left(\frac{PH_{t+3}}{1 - PH_{t+3}}\right), \sigma_{proc}\right).$$

Note that the preceding three expressions link S_{-t+3} to S_{-t} . This is the mechanistic model connecting the true spawner abundance time-series together. The last step is to model the observation of spawner abundance as the result of normally-distributed observational error centered on the true but latent spawner abundance:

$$S_t \sim \text{Normal}(\underline{S}_t, \sigma_{T,obs}).$$

The subscript T indicates that there are different possible magnitudes of observation errors for the $T = 1, 2, 3$ time periods when abundance was estimated with different sampling designs.

2.7. Hierarchical modeling

Barrowman et al. (2003) advocate abandoning the pretense that ecological parameters among populations are unrelated, and suggest that such parameters should instead be treated as coming from a common distribution. This gives rise to "hyperparameters" of statistical distributions from which ecological (recruitment) parameters are derived. Data sets from multiple populations can thus be combined together and analyzed in a single, meta-analytic framework. In a Bayesian context, this can be accomplished with hierarchical modeling (Liermann and Hilborn, 1997; Barrowman et al., 2003; Michielsens and McAllister, 2004).

The three parameters of the logistic hockey stick model are treated hierarchically. That is, they are assumed to come from a parameter-generating distribution that itself is defined by hyperparameters. The parameters of the logistic hockey-stick function for each population (p) are assumed to come from the following distributions:

$$\alpha_p \sim \text{Normal}(\mu_\alpha, \sigma_\alpha)$$

$$\theta_p \sim \text{Normal}(\mu_\theta, \sigma_\theta)$$

$$\gamma_{p,t=1970:1998} \sim \text{Normal}(\mu_{\gamma_1}, \sigma_\gamma)$$

$$\gamma_{p,t=1999:2014} \sim \text{Normal}(\mu_{\gamma_2}, \sigma_\gamma).$$

Note that the parameter controlling the location of the recruitment inflection point (γ) is assumed to come from two distributions with means that are potentially different during the period 1970 to 1998 and 1999 to 2015. This is what allows us to detect whether freshwater recruitment has changed after the implementation of biologically conservative actions. The prior distributions for other parameters and hyperparameters are given in Table 2. Note that the difference in scale between process variation and observation error reflects the fact that process variation is lognormal whereas observation error is normally distributed.

We fit three hierarchical state-space models to our data (Table 3). All three models allowed change in the smolt recruitment functions up to and after 1998. Models differ with respect to whether or not they allow hatchery-origin spawning fish to have lower reproductive success than their wild-origin counterparts (Ψ), and whether or not a quadratic function of freshwater streamflow was included. The purpose of the three models was to determine the sensitivity of estimates of change in smolt recruitment to model specification. The JAGS code for Model A is included in Supplement 1 and the corresponding data are included in

Table 2
Prior distributions used in the most complex hierarchical state-space model (Model A) of Oregon coastal coho population dynamics.

Parameter	Distribution
μ_α	Uniform(10, 250)
σ_α	Uniform(20, 100)
μ_0	Uniform(0, 5)
σ_0	Uniform(0, 2)
μ_{y1}	Uniform(200, 2000)
μ_{y2}	Uniform(200, 2000)
σ_y	Uniform(100, 1000)
σ_{proc}	Uniform(0, 2)
$\sigma_{obs,t=1974:1990}$	Uniform(1, 1000)
$\sigma_{obs,t=1991:1998}$	Uniform(1, 1000)
$\sigma_{obs,t=1999:2015}$	Uniform(1, 1000)
β_1	Uniform(−5, 5)
β_2	Uniform(−5, 5)
ψ	Beta(1, 1)

Table 3

Three models of Oregon coastal coho population dynamics (columns) are distinguished from one another by whether or not (i) the relative reproductive success (RRS) of hatchery-origin fish is estimated, and (ii) a quadratic function of freshwater stream flow is used as a covariate in smolt recruitment. All three models permitted change in the mean, across-population inflection point in coastal coho smolt recruitment before (μ_{y1}) and after (μ_{y2}) 1998. The probability (Pr) that smolt recruitment declined was found by conducting random permutations of the posterior distributions. The odds ratio for improved recruitment is computed directly from the associated probabilities.

Model ID	A	B	C
Change recruitment?	Yes	Yes	Yes
Estimate RRS?	Yes	No	No
Include Flow?	Yes	Yes	No
μ_{y1}	623	657	656
μ_{y2}	407	434	434
$\text{Pr}[\mu_{y1} > \mu_{y2}]$	0.87	0.86	0.86
Odds ratio $\mu_{y2} > \mu_{y1}$	1 in 6.8	1 in 6.2	1 in 6.1

Supplement 2.

2.8. Model fitting and diagnostics

We ran 4 Markov chain Monte Carlo (MCMC) simulations per model using JAGS software (Plummer, 2003). The JAGS software ran the four MCMC simulations in a parallel process that was implemented within MATLAB (MATLAB, 2015) using “Matjags.m”. All chains were allowed to burn-in for 150,000 iterations. We thinned to 1:13 samples, and retained a total of 125,000 samples per chain. The autocorrelation function of the (thinned) samples showed insignificant correlations at the fifth lag. We looked at the estimated potential scale reduction values computed by JAGS to diagnose model convergence (Gelman and Rubin, 1992). We also visually inspected trace plots of the MCMC output for signs of good mixing and convergence (Link and Barker, 2010). We computed MCMC error for parameters using a “batch means” approach (Roberts, 1996). To do this, each MCMC chain, $\{X_i\}$, of length $I = 125,000$, was broken into K batches of length $b = \sqrt{I}$, and then for each batch we computed the sample mean.

$$Y_k = \frac{\sum_{i=(k-1)b+1}^{kb} X_i}{b}.$$

We then computed the K batches.

$$se_{MCMC} = \sqrt{\frac{\frac{b}{K-1} \sum_{k=1}^K (Y_k - \hat{\mu})^2}{b}},$$

where $\hat{\mu}$ is the sample mean of the entire MCMC simulation. We took the mean of standard errors associated with each parameter's four MCMC simulations after noting that they were extremely similar.

We assessed model goodness of fit using a Bayesian posterior predictive check (Gelman et al., 1996). For all years (t) and populations (p) a chi-square discrepancy test statistic was computed for Model A: $\chi_{t,p}^2 = \frac{(S_{t,p} - S)^2}{S}$. Simulated values $y \sim \text{Normal}(S, \sigma_{T,obs})$ were created within the MCMC and used to compute simulated chi-square statistics: $\chi \cdot \text{sim}_{t,p}^2 = \frac{(y \cdot \text{sim}_{t,p} - S)^2}{S}$. An omnibus statistic for goodness of fit was created by summing $\chi_{t,p}^2$ and $\chi \cdot \text{sim}_{t,p}^2$ across time (t) and then comparing the frequency with which $\chi \cdot \text{sim}_p^2 > \chi_p^2$. This is also known as a Bayesian p -value (Kéry and Royle, 2016).

3. Results

The estimated potential scale reduction values for all parameter within all models were good (1 ± 0.01), and almost all were excellent (1 ± 0.001). The goodness of fit statistic for Model A was 0.52, suggesting that there was no evidence for lack-of-fit. Visual inspection of trace plots revealed that the MCMC samples were well mixed and stable across iterations. We monitored the node S within the MCMC and then verified that none of our distributional assumptions allowed S to become negative.

All three models allow smolt recruitment dynamics to change before and after 1998, which broadly correspond to pre and post-recovery-action periods. For Model A, the mean of the across-population recruitment inflection point prior to 1998 (μ_{y1}) is 623, but after 1998 the value (μ_{y2}) is 407 (Table 3). This means that the spawner abundance at which adult recruitment begins to asymptote is lower in the latter period than the earlier period. Random draws from the posterior distributions of μ_{y1} and μ_{y2} (Fig. 4a) show that there is a 0.87 probability that recruitment was worse after 1998 than before 1998. This is equivalent to an odds ratio of 1 in 6.8 that there was improvement in recruitment between these periods. Models B and C yielded estimates of μ_{y1} and μ_{y2} that are similar to Model A (Table 3), indicating that model choice about including streamflow or estimating the relative reproductive success of hatchery fish did not affect the main result. At the scale of individual populations, point estimates for the smolt recruitment inflection point were lower in all 18 populations prior to 1998 than after 1998, though none of the differences in point estimates within populations was statistically significant (Fig. 4b).

The relative reproductive success of hatchery-origin fish in Model A was 0.51, with 95% highest probability density intervals extending from 0.19 to 0.89 (Fig. 5). The point estimate of relative reproductive success is very similar to the mean of the prior distribution (0.5), so we re-estimated this parameter using a Beta ($a = 2, b = 1$) prior distribution, which has mean equal to $a / (a + b) = 0.66$. This informative prior caused the point estimate of the relative reproductive success to increase to 0.57.

There were several relatively high and low stream flow events experienced by brood years 1970–1976, and then again in the late 1990s (Fig. 1d). Stream flow was near average for brood years 1983–1993, and again in the 2000s (Fig. 1d). We did not find statistically significant effects of stream flow on smolt production in any of the models (Supplement 3). Observation error (σ_{obs}) was similar from 1991 to 1998, and 1999–2014. However, observation error during the earliest period, 1974–1990, was 195% greater than the average of the latter two periods, with variation across models (Supplement 3).

4. Discussion

After accounting for harvest, variable ocean survival, and poor reproductive success of hatchery-origin fish spawning in the wild, we found no evidence for improvement in the production of coastal coho smolts per spawner after 1998. Thus, recent increases in adult

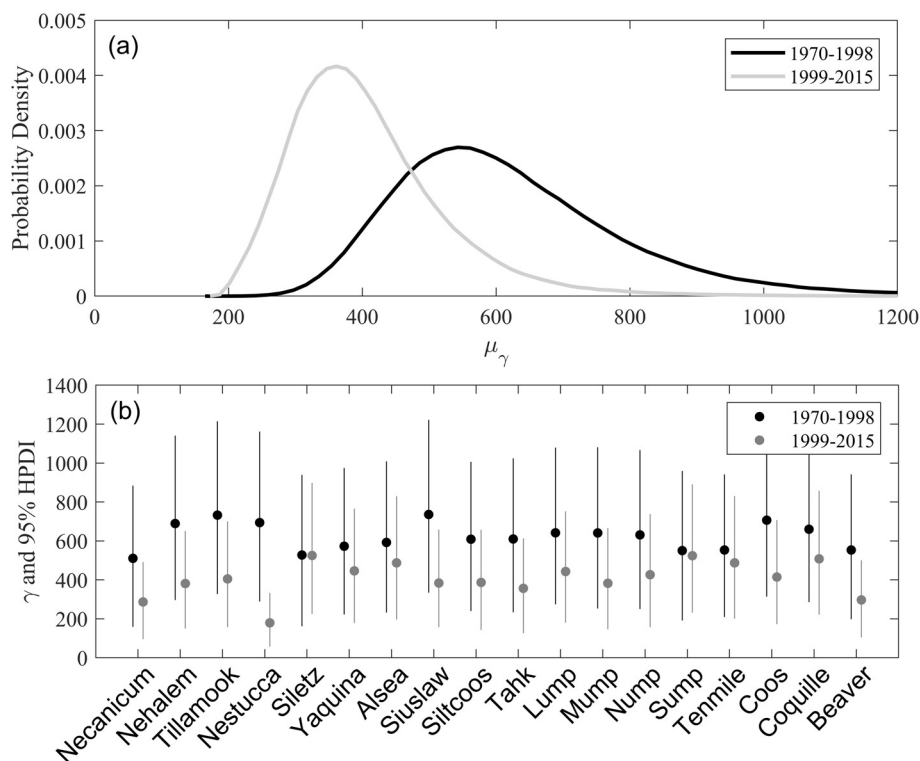


Fig. 4. Posterior probability distribution of the mean, all-population distribution of inflection points (γ) in logistic hockey stick smolt recruitment functions from 1970 to 1998 and 1999 to 2015 (a). Individual population point estimates and 95% highest probability density intervals for inflection points (γ) in logistic hockey stick smolt recruitment functions from 1970 to 1998 and 1999 to 2015 (b). Model A used in both (a) and (b).

abundance can be attributed to improved ocean survival, not improved fresh water conditions. Although our results suggest that there has not been a positive change in the production of smolts per spawner, many of the conservation efforts enacted since the late 1990s are not strictly expected to have direct and immediate effects on smolts per spawner. For example, an obvious effect of reducing harvest is that more adults escape the fishery and reproduce. This could improve smolts per spawner at low spawner abundances by reducing the frequency of

negative density-dependent “Allee” effects (Courchamp et al., 1999). Our recruitment models were designed to detect such depensation (Liermann and Hilborn, 1997; Barrowman et al., 2003). Reducing harvest may also generally increase the genetic and life-history diversity of a population (Thériault et al., 2008; Hard et al., 2008), thereby increasing density-dependent smolt production through evolutionary niche expansion (Holt et al., 2004; Faulks et al., 2015). The time required for evolutionary effects to unfold, which includes not

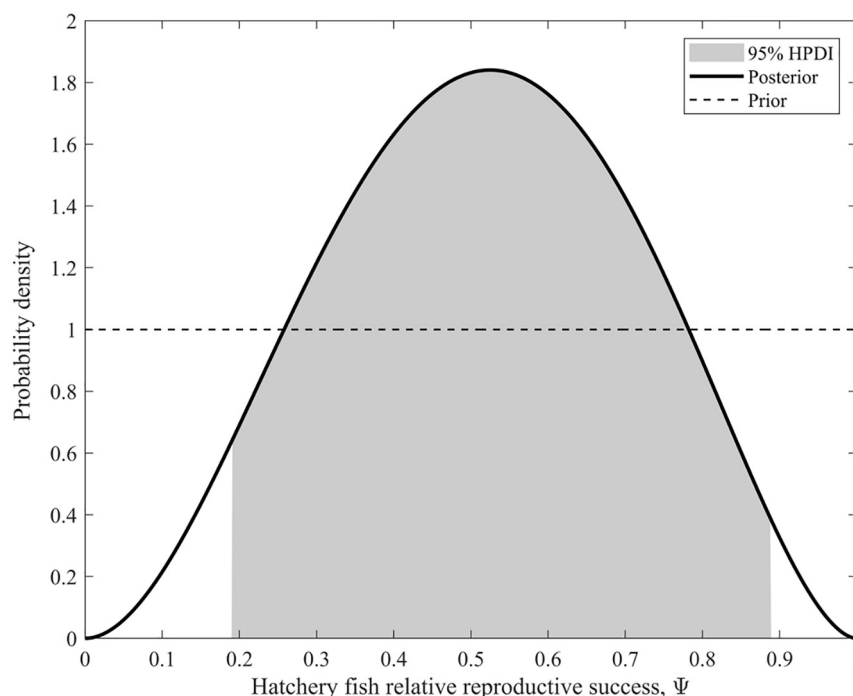


Fig. 5. Posterior probability density of hatchery fish reproductive success relative to natural-origin fish (solid line). Shaded area is the 95% highest probability density interval [0.19, 0.89]. The prior distribution used in the analysis is equi-probable over the entire range of possible values for hatchery fish relative reproductive success (dashed line).

only the effects of fishing moratoria on population growth rate (Dunlop et al., 2015), but also reversing the fitness effects of domestication selection caused by gene flow from hatchery-reared fish into the naturally reproducing population (Ford, 2002), is generally longer than the length of time analyzed here (Anderson and Brander, 2009). However, there is growing evidence for rapid evolution (Carroll et al., 2007; Schoener, 2011; Kupperman et al., 2016), so further study is needed to understand the rate of change in adaptive genetic variation (Steiner et al., 2013) of coastal coho.

Our primary finding, that coastal coho smolts per spawner has not increased, is sensitive to the modeled estimates of ocean survival before 1998. If the estimates of ocean survival prior to 1998 are biased low, then there could be an improvement in freshwater recruitment that went undetected. The GAM model relating ocean survival from 1998 to 2015 to oceanographic variables produces a good fit, and variables were selected based on the predictive performance of adult abundance dating back to 1970 (Rupp et al., 2012b). Although the variables were selected from a large collection of potential predictor variables, the methods used by Rupp et al. (2012b) to perform variable selection should reduce spuriousness. Nonetheless, it is possible that the GAM curvatures relating oceanographic variables to survival change through time as the values of other, unmeasured oceanographic variables change (Yasunaka et al., 2016). This kind of ecological complexity sets an upper limit to what can be usefully known (Ward et al., 2014; Schindler and Hilborn, 2015), but improvements are certainly possible (Petchy et al., 2015). Additional model checking is needed as more data become available (Myers, 1998).

Increasing proportions of hatchery-origin coho salmon on spawning grounds have been associated with reduced density-dependent production of adults (Buhle et al., 2009) and population intrinsic productivity (Chilcote et al., 2011). If hatchery-origin spawners have poor reproductive success in the wild (Fleming and Gross, 1993; Araki et al., 2007; Christie et al., 2012), then we should expect the spawner-recruit relationship of a mixed-origin population (hatchery and natural) to be lower than an equivalent all-natural population. Here, we used a novel demographic method to estimate the relative reproductive success of hatchery-origin spawners so that we could remove the effects of temporal variation in the proportion of hatchery-origin spawners on recruitment performance. Our point estimate of the relative reproductive success of hatchery-origin fish (0.51) is similar to findings from genetic analyses (Christie et al., 2014). Uncertainty in our demographically-derived estimate of relative reproductive success may be partially attributable to differences in the relative frequency of hatchery-X-natural and hatchery-X-hatchery mating that occurs over years and populations because of different hatchery-fish rearing and release strategies.

It is possible that any improvement in smolt production from the reductions in hatchery origin fish on the spawning grounds and relief from fishing-induced evolution has been offset by freshwater habitat degradation. However, salmonid freshwater habitat metrics on the Oregon coast show mixed trends (increasing, decreasing, no change) over 1998–2007, depending the specific habitat features and latitudinal position on the Oregon coast (Anlauf et al., 2011). Similar analyses using more recent data produce similar results (Anlauf-Dunn, personal communication). There is uncertainty about how these freshwater habitat metrics are related to fish fitness, but our finding that coastal coho smolt production has not increased seems broadly consistent with trends described by Anlauf et al. (2011). Habitat management activities have only recently focused on restoring watershed processes that create high-quality freshwater habitat (ODFW, 2007). The time required to reverse decades of habitat destruction that at times affected geological processes (e.g. “splash dams” were used to raise water levels and then drive logs downstream in events that would scour streams to bedrock) likely extends well beyond the time frame analyzed here. Future work should examine which freshwater habitat factors continue to decline, how freshwater habitat features interactively relate to fish fitness, and what further management actions might be needed to improve

freshwater productivity for coho salmon.

Although coho spawner abundances on the Oregon Coast have been estimated with sophisticated probabilistic sampling designs since the late 1990s, uncertainty in the estimates of spawner population abundance is not negligible. State-space models are particularly well suited for these data because they explicitly partition process and observation error. The models used here allow different magnitudes of observational error throughout three periods when different sampling designs were used. Our results are consistent with the expectation that more recent sampling methods are more precise. This is expected because index sites (Falcu et al., 2016) used to estimate abundance in the early part of these time series should have low precision when density-dependent habitat selection occurs (Falcu, 2015). Separating observation error from process noise could be useful when performing population viability analysis because a population simulation should include process variation but not the observational error associated with parameter estimation. State-space modeling can thus improve viability analysis by yielding estimates of uncertainty that pertain only to the biological process (White, 2000; Hinrichsen, 2009; Dennis and Ponciano, 2014). Our fitted model could be used to simulate extinction probabilities when ocean conditions that are beyond management control return to low values observed during the mid-1990s.

There is uncertainty about the details of compensatory density-dependence processes that influence coho recruitment to the smolt stage. A large part of that uncertainty is due to the interaction of some of the underlying mechanisms that drive the rate and direction of these processes. Compensation rates and positive time trends in population productivity are affected by both the magnitude of management effects and the time scales over which responses occur. These processes occur along with uncontrollable environmental variability. Despite this uncertainty, the management strategy to improve Oregon coastal coho productivity has been prudent and multipronged. For example, (i) virtually eliminating harvest has ensured the maximum number of fish escape to spawn, (ii) drastically reducing the number of hatchery fish on the spawning grounds minimizes the risks of hatchery fish interacting with wild fish, and (iii) efforts to restore freshwater habitat conditions have been consistent with an understanding of life-cycle habitat requirements for coho (ODFW, 2007). Fish managers will need to remain vigilant, collaborative, and adaptive if monitoring data suggest that modified or additional strategies should be implemented.

Concern about climate change, ecosystem regime shifts, and the status of global fisheries begs questions about how managers should respond. Measuring the effectiveness of management actions is challenging because potentially confounding factors usually cannot be controlled, and the mechanisms linking actions to outcomes may be poorly understood. However, understanding management effectiveness can alleviate reliance on the precautionary principle as a hedge against uncertainty (Falcu, 2016), and will hopefully lead to cost-effective management actions that achieves society's goals. Thus understanding management regime shifts on population dynamics is no less important than understanding naturally-caused ecosystem regime shifts. Tools for detecting nonstationary population dynamics can be applied to both kinds of regime shifts.

We developed a novel and relatively complex model to integrate numerous sources of data that span many decades and many geographically distributed populations. We did this in a way that enabled us to tease apart potentially confounding factors affecting Oregon coast coho. Researchers and conservation managers of other systems will need to think creatively and critically about how to tailor models to their data and questions. Difficult problems are not likely to have simple answers. Custom modeling, not the rote application of existing statistical routines or packages, must be paired with careful interpretation in order to advance our understanding of complex conservation problems.

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References

- Anderson, K.H., Brander, K., 2009. Expected rate of fisheries-induced evolution is slow. *Proc. Natl. Acad. Sci. U. S. A.* 106, 11657–11660.
- Anlauf, K.J., Gaeuman, W., Jones, K.K., 2011. Detection of regional trends in salmon habitat in coastal streams, Oregon. *Trans. Am. Fish. Soc.* 140, 52–66.
- Araki, H., Cooper, B., Blouin, M.S., 2007. Genetic effects of captive breeding cause a rapid, cumulative fitness decline in the wild. *Science* 318, 100–103.
- Barrowman, N.J., Myers, R.A., 2000. Still more spawner-recruitment curves: the hockey stick and its generalizations. *Can. J. Fish. Aquat. Sci.* 57, 665–676.
- Barrowman, N.J., Myers, R.A., Hilborn, R., Kehler, D.G., Field, C.A., 2003. The variability among populations of coho salmon in the maximum reproductive rate and depensation. *Ecol. Appl.* 13, 784–793.
- Bradford, M.J., Taylor, G.C., Allan, J.A., 1997. Empirical review of coho salmon smolt abundance and the prediction of smolt production at the regional level. *Trans. Am. Fish. Soc.* 126, 49–64.
- Buhle, E.R., Holsman, K.K., Scheurell, M.D., Albagh, A., 2009. Using an unplanned experiment to evaluate the effects of hatcheries and environmental variation on threatened populations of wild salmon. *Biol. Conserv.* 142, 2449–2455.
- Carroll, S.P., Hendry, A.P., Reznick, D.N., Fox, C.W., 2007. Evolution on ecological time-scales. *Funct. Ecol.* 21, 387–393.
- Chilcote, M.W., Goodson, K.W., Falcý, M.R., 2011. Reduced recruitment performance in natural populations of anadromous salmonids associated with hatchery-reared fish. *Can. J. Fish. Aquat. Sci.* 68, 511–552.
- Christie, M.R., Marine, M.L., French, R.A., Blouin, M.S., 2012. Genetic adaptation to captivity can occur in a single generation. *Proc. Natl. Acad. Sci. U. S. A.* 109, 238–242.
- Christie, M.R., Ford, M.J., Blouin, M.S., 2014. On the reproductive success of early-generation hatchery fish in the wild. *Evol. Appl.* 7, 883–896.
- Courchamp, F., Clutton-Brock, T., Grenfell, B., 1999. Inverse density dependence and the Allee effect. *Trends Ecol. Evol.* 14, 405–409.
- Dennis, B., Ponciano, J.M., 2014. Density-dependent state-space model for population-abundance data with unequal time intervals. *Ecology* 95, 2069–2076.
- Dunlop, E.S., Eikeset, A.M., Stenseth, N.C., 2015. From genes to populations: how fisheries-induced evolution alters stock productivity. *Ecol. Appl.* 25, 1860–1868.
- Falcý, M.R., 2015. Density-dependent habitat selection of spawning Chinook salmon: broad-scale evidence and implications. *J. Anim. Ecol.* 84, 545–553.
- Falcý, M.R., 2016. Conservation decision making: integrating the precautionary principle with uncertainty. *Front. Ecol. Environ.* 14, 499–504.
- Falcý, M.R., McCormick, J.L., Miller, S.A., 2016. Proxies in practice: calibration and validation of multiple indices of animal abundance. *J. Fish and Wildl. Manag.* 7, 117–128.
- Faulks, L., Svanbäck, R., Ragnarsson-Stabo, H., Eklöv, P., Östman, Ö., 2015. Intraspecific niche variation drives abundance-occupancy relationship in freshwater fish communities. *Am. Nat.* 186, 272–283.
- Fleming, I.A., Gross, M.R., 1993. Breeding success of hatchery and wild coho salmon (*Oncorhynchus kisutch*) in competition. *Ecol. Appl.* 3, 230–245.
- Ford, M., 2002. Selection in captivity during supportive breeding may reduce fitness in the wild. *Conserv. Biol.* 16, 815–825.
- Gelman, A., Hill, J., 2006. *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge University Press, Cambridge.
- Gelman, A., Rubin, D., 1992. Inference from iterative simulation using multiple sequences. *Stat. Sci.* 7, 457–511.
- Gelman, A., Meng, X.-L., Stern, H.S., 1996. Posterior predictive assessment of model fitness via realized discrepancies. *Stat. Sin.* 6, 733–807.
- Hard, J.J., Gross, M.R., Heino, M., Hilborn, R., Kope, R.G., Law, R., Reynolds, J.D., 2008. Evolutionary consequences of fishing and their implications for salmon. *Evol. Appl.* 1, 388–408.
- Hilborn, R., Walters, C.J., 1992. *Quantitative Fisheries Stock Assessment: Choice, Dynamics, and Uncertainty*. Springer and Routledge, Chapman & Hall.
- Hinrichsen, R.A., 2009. Population viability analysis for several populations using multivariate state-space models. *Ecol. Model.* 220, 1197–1202.
- Holt, R.D., Knight, T.M., Barfield, M., 2004. Allee effects, immigration, and the evolution of species' niches. *Am. Nat.* 163, 253–262.
- Jacobs, S.E., Nickelson, T.E., 1998. Use of stratified random sampling to estimate the abundance of Oregon coastal Coho Salmon. Oregon Department of Fish and Wildlife, Report F-145-R-09, Oregon, Salem.
- Kéry, M., Royle, J.A., 2016. *Applied Hierarchical Modeling in Ecology: Analysis of Distribution, Abundance and Species Richness in R and BUGS*. Vol. 1 Academic Press, London.
- Kilduff, D.P., Di Lorenzo, E., Botsford, L.W., Teo, S.L.H., 2015. Changing central Pacific El Niños reduce stability of North American salmon survival rates. *Proc. Natl. Acad. Sci. U. S. A.* 112, 10962–10966.
- King, J.R., McFarlane, G.A., Punt, A.E., 2015. Shifts in fisheries management: adapting to regime shifts. *Philos. Trans. R. Soc. B* 370, 20130277.
- Kuparinen, A., Hutchings, J.A., Waples, R.S., 2016. Harvest-induced evolution and effective population size. *Evol. Appl.* 9, 658–672.
- Lawson, P.W., Logerwell, E.A., Mantua, N.J., Francis, R.C., Agostini, V.N., 2004. Environmental factors influencing freshwater survival and smolt production in Pacific Northwest coho salmon (*Oncorhynchus kisutch*). *Can. J. Fish. Aquat. Sci.* 61, 360–373.
- Lewis, M., Brown, E., Sounhein, B., Weeber, M., Suring, E., Truemper, H., 2009. Status of Oregon stocks of coho salmon, 2004 through 2008. In: *Monitoring Program Report Number OPSW-ODFW-2009-3*. Oregon Department of Fish and Wildlife, Salem, Oregon.
- Liermann, M., Hilborn, R., 1997. Depensation in fish stocks: a hierarchic Bayesian meta-analysis. *Can. J. Fish. Aquat. Sci.* 54, 1976–1984.
- Link, W.A., Barker, R.J., 2010. *Bayesian Inference with Ecological Applications*. Academic Press, London.
- Lonergan, M., 2014. Data availability constrains model complexity, generality, and utility: a response to Evans et al. *Trends Ecol. Evol.* 29, 301–302.
- Mallet, J., 2012. The struggle for existence: how the notion of carrying capacity, K, obscures the links between demography, Darwinian evolution, and speciation. *Evol. Ecol. Res.* 14, 627–665.
- MATLAB Release, 2015. The MathWorks, Inc., Natick, Massachusetts, United States. <https://www.mathworks.com/products/matlab.html>.
- Melcher, C., 2005. Oregon Coastal Coho Assessment Part 4C: Oregon Coast Natural Coho Harvest Management and In-river Salmon Fisheries. Oregon Department of Fish and Wildlife Report. pp. 1–23.
- Meyer, R., Millar, R.B., 1999. BUGS in Bayesian stock assessment. *Can. J. Fish. Aquat. Sci.* 56, 1078–1086.
- Michielsens, C.G.J., McAllister, M., 2004. A Bayesian hierarchical analysis of stock-recruit data: quantifying structural and parameter uncertainties. *Can. J. Fish. Aquat. Sci.* 61, 1032–1047.
- Millar, R.B., Meyer, R., 2000a. Non-linear state space modelling of fisheries biomass dynamics by using Metropolis-Hastings within Biggs sampling. *Appl. Stat.* 49, 327–342.
- Millar, R.B., Meyer, R., 2000b. Bayesian state-space modeling of age-structured data: fitting a model is just the beginning. *Can. J. Fish. Aquat. Sci.* 57, 43–50.
- Myers, R.A., 1998. When do environment-recruitment correlations work? *Rev. Fish Biol. Fish.* 8, 285–305.
- Nicholas, J., McIntosh, B., Bowles, E., 2005. Oregon Coastal Coho Assessment Part 1: Synthesis. Oregon Department of Fish and Wildlife Report. pp. 1–69.
- ODFW Oregon Department of Fish and Wildlife, 2007. *Oregon Coast Coho Conservation Plan for the State of Oregon*. http://www.dfw.state.or.us/fish/crp/docs/coastal_coho/final/Coho_Plan.pdf.
- ODFW, 2013. Technical Revision to the OCN Coho Work Group Harvest Matrix. <https://nrimp.dfw.state.or.us/CRL/reports/SLCMP/A13TechnicalRevision2013.pdf>.
- OWEB Oregon Watershed Enhancement Board, 2017. Investment Tracking Tool. <http://www.oregon.gov/OWEB/docs/oitt.html>, Accessed date: 20 March 2017.
- Petchy, O.L., Pontarp, M., Massie, T.M., et al., 2015. The ecological forecast horizon, and examples of its uses and determinants. *Ecol. Lett.* 18, 597–611.
- Pinsky, M.L., Worm, B., Fogarty, M.J., Sarmiento, J.L., Levin, S.A., 2013. Marine taxa track local climate velocities. *Science* 341, 1239–1242.
- Plummer, M., 2003. JAGS: a program for analysis of Bayesian graphical models using Gibbs sampling. In: *Proceedings of the 3rd International Workshop on Distributed Statistical Computing*, Vienna, March 20–22, 2003.
- Punt, A.E., Butterworth, D.S., de Moor, C.L., De Oliveira, J.A.A., Haddon, M., 2016. Management strategy evaluation: best practices. *Fish. Fish.* 17, 303–334.
- Rivot, E., Prévost, E., Parent, E., Baglinière, J.L., 2004. A Bayesian state-space modelling framework for fitting a salmon state-structured population dynamic model to multiple time series of field data. *Ecol. Model.* 179, 463–485.
- Roberts, G.O., 1996. Markov chain concepts related to sampling algorithms. In: Gilks, W.R., Richardson, S., Spiegelhalter, D.J. (Eds.), *Markov Chain Monte Carlo in Practice*. Chapman and Hall, London, pp. 45–57.
- Rupp, D.E., Wainwright, T.C., Lawson, P.W., 2012a. Effect of forecast skill on management of the Oregon coast coho salmon (*Oncorhynchus kisutch*) fishery. *Can. J. Fish. Aquat. Sci.* 69, 1016–1032.
- Rupp, D.E., Wainwright, T.C., Lawson, P.W., Peterson, W.T., 2012b. Marine environment-based forecasting of coho salmon (*Oncorhynchus kisutch*) adult recruitment. *Fish. Oceanogr.* 21, 1–19.
- Salmon Technical Team, S.T.T., 1995. Stock abundance analysis for 1995 ocean salmon fisheries: preseason report I. In: *Pacific Fisheries Management Council Report*, pp. 1–99.
- Schindler, D.E., Hilborn, R., 2015. Prediction, precaution, and policy under global change. *Science* 347, 953–954.
- Schoener, T.W., 2011. The newest synthesis: understanding the interplay of evolutionary and ecological dynamics. *Science* 331, 426–429.
- Steiner, C.C., Putnam, A.S., Hoeck, P.E.A., Ryder, O.A., 2013. Conservation genomics of threatened animal species. *Ann. Rev. Anim. Biosci.* 1, 261–281.
- Su, Z., Peterman, R.M., 2012. Performance of a Bayesian state-space model of semelparous species for stock-recruitment data subject to measurement error. *Ecol. Model.* 224, 76–89.
- Suring, E., Burns, P., Constable Jr., R.J., Lorion, C.M., Wiley, D.J., 2015. Salmon life cycle monitoring in Western Oregon streams, 2012–2014. In: *Oregon Department of Fish and Wildlife Monitoring Program Report OPSW-ODFW-2015-2*.
- Szuwalski, C.S., 2016. Changing fisheries productivity and food security. *Proc. Natl. Acad. Sci. U. S. A.* 113, E1773–E1774.
- Szuwalski, C.S., Vert-Pre, K.A., Punt, A.E., Branch, T.A., Hilborn, R., 2015. Examining

- common assumptions about recruitment: a meta-analysis of recruitment dynamics for worldwide marine fisheries. *Fish Fish.* 16, 633–648.
- Thériault, V., Dunlop, E.S., Dieckmann, U., Bernatchez, L., Dodson, J., 2008. The impact of fishing-induced mortality on the evolution of alternative life-history tactics in brook charr. *Evol. Appl.* 1, 409–423.
- Vert-pré, K.A., Amoroso, R.O., Jensen, O.P., Hilborn, R., 2013. Frequency and intensity of productivity regime shifts in marine stocks. In: *Proceedings of the National Academy of Sciences*. Vol. 110. pp. 1779–1784.
- Ward, E.J., Holmes, E.E., Thorson, J.T., Colen, B., 2014. Complexity is costly: a meta-analysis of parametric and non-parametric methods for short-term population forecasting. *Oikos* 123, 652–661.
- White, G.C., 2000. Population viability analysis: data requirements and essential analyses. In: Boitani, L., Fuller, T.K. (Eds.), *Research Techniques in Animal Ecology: Controversies and Consequences*. Columbia University Press, New York.
- Worm, B., Hilborn, R., Baum, J.K., et al., 2009. Rebuilding global fisheries. *Science* 325, 578–585.
- Yasunaka, S., Ono, T., Nojiri, Y., Whitney, F.A., Wada, C., Murata, A., Nakaoka, S., Hosoda, S., 2016. Long-term variability of surface nutrient concentrations in the North Pacific. *Geophys. Res. Lett.* 43, 3389–3397.